

LINEAR DISCRIMINANT ANALYSIS

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Agenda

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- Recap of PCA
 - ▣ Major Limitations
- Dimensionality Reduction in Classification
 - ▣ Key Requirements
- Let's see how PCA fails!
- Linear Discriminant Analysis (LDA)
- PCA versus LDA

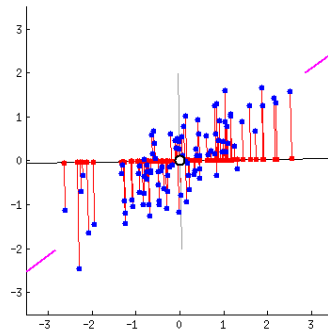
LDA

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PCA: A Quick Recap

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- A dimensionality reduction technique
 - ▣ Finds the most **accurate lower dimensional** representation based on **maximum variance directions**



LDA

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PCA: A Quick Recap

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- Key Features
 - ▣ **Unsupervised** (does not utilize labels)
 - ▣ Focuses on **variance** (not always useful or relevant)

LDA

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PCA: Limitations

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- Limitations
- Not suitable for discrimination/classification
 - ▣ Does not take **class information** into consideration
 - ▣ PCA is an **optimal** dimensionality-reduction technique for **data representation**, not for discrimination/classification problems

LDA

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Dimensionality Reduction in Classification



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- A simple real-world classification example: Identifying species of a fish on a conveyor belt
 - ▣ Species: **Sea bass and salmon**

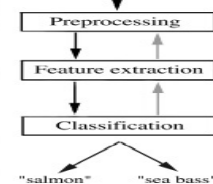


Image source: Pattern Classification by Duda, Hart and Stork

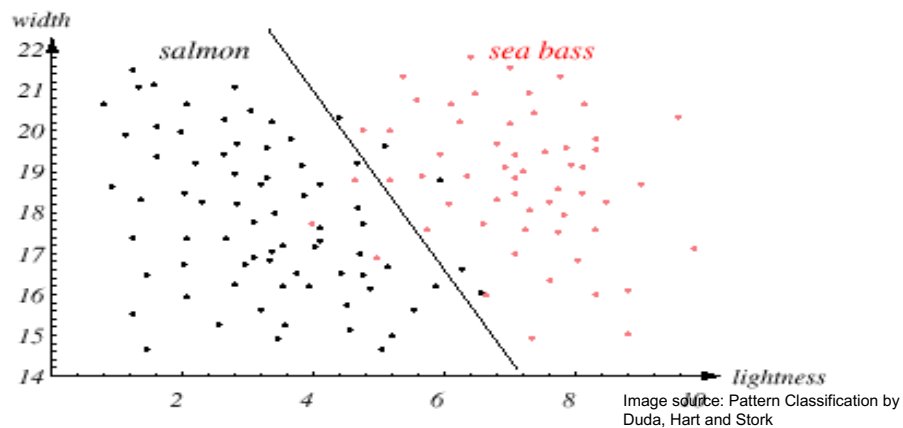
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Dimensionality Reduction in Classification



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- Two features for classification



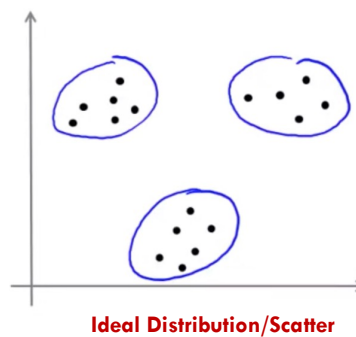
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Dimensionality Reduction in Classification



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- Key Requirements for Classification
 - Each class is as **tight** (compact) as possible
 - Their centroids are as **far** from each other as possible



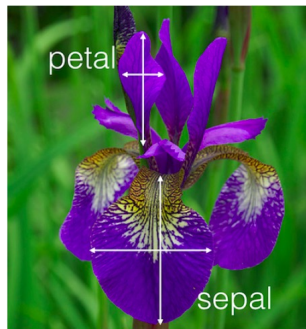
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Let's see how PCA Fails!



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- PCA is not an effective **dimensionality technique for classification**
- The **Iris Dataset**



	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

One of the earliest known datasets used for evaluating classification methods, widely used in ML

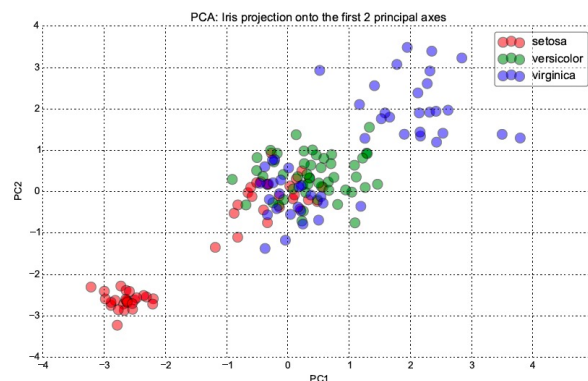
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Let's see how PCA Fails!



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- **Problem:** Given a new measurement of these features, predict the Iris species based on a projection onto a low-dimensional space



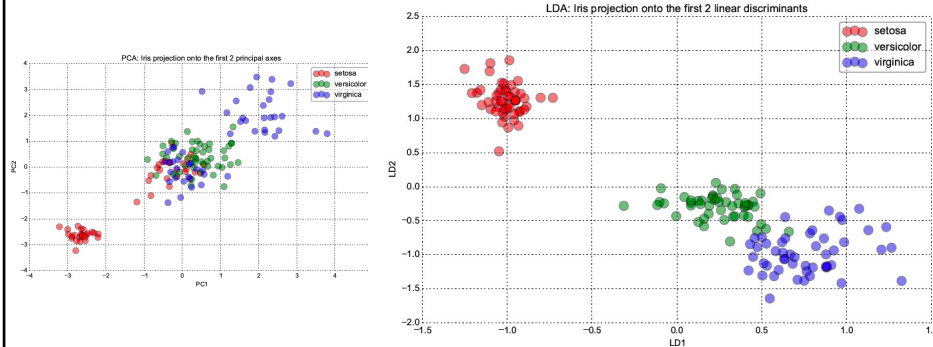
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Let's see how PCA Fails!



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- **Conclusion:** PCA may not be ideal to separate the classes well

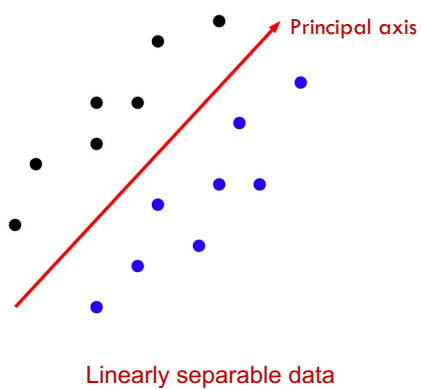


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Why PCA Fails



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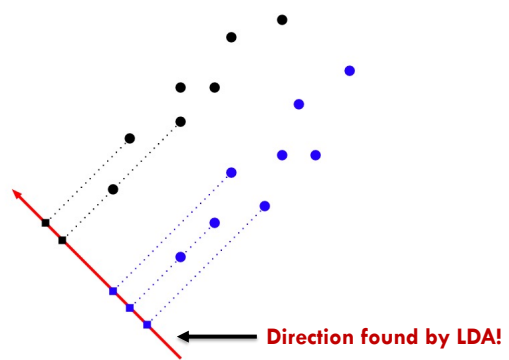
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Linear Discriminant Analysis (LDA)



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- Solution: A DR technique that tries to **preserve the discriminatory information** between different classes of the data set



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Linear Discriminant Analysis



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- Key Idea
 - Select the **best direction for projection** that
 - **Maximizes** their **between-class separability** while
 - **Minimizing** their **within-class variability**
- Key Requirements for Classification

- Each class is as **tight** (compact) as possible
 - Their centroids are as **far** from each other as possible

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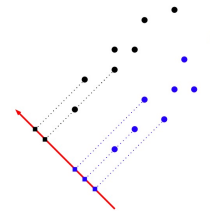
Linear Discriminant Analysis



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Select the **best direction for projection** that

- Maximizes their between-class separability while
- Minimizing their within-class variability
- How do we **quantify the separation** between the two classes?
 - Measure the distance between the two class means in the projected space!



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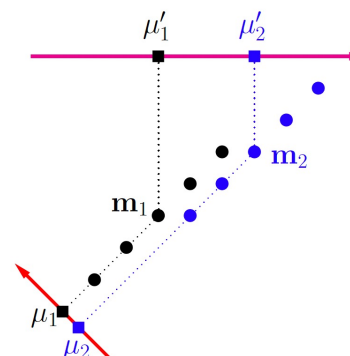
Linear Discriminant Analysis



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How do we **quantify the separation** between the two classes?

- Measure the distance between the two class means in the projected space!
- This criterion alone might fail!



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Linear Discriminant Analysis



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How do we **quantify the within-class variability**?

- Compute variances of the projected classes

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Linear Discriminant Analysis

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- Linear Discriminant Analysis/ Fisher Discriminant Analysis
 - ▣ Objective: Perform dimensionality reduction while preserving as much of the class discriminatory information as possible.
 - Directions along which classes are best separated
- Find a projection W that
 - ▣ Maximizes the **between-class** separability
 - ▣ Minimize **within-class** variability
- Maximize **Fisher's criteria**

LDA

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Linear Discriminant Analysis

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□ Fisher's criteria

$$J(w) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

Measure of **between-class separability**

Total **within-class variance/scatter**

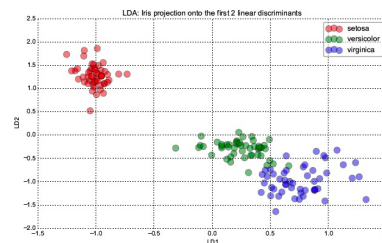
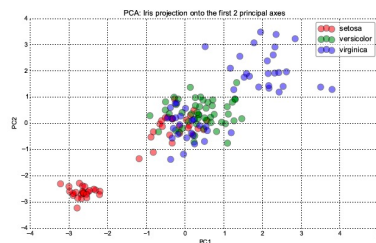
LDA

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PCA versus LDA

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- Like PCA, LDA looks at an **eigenvalue problem** for **dimensionality reduction**
- LDA: **Eigenvalues** indicate the **importance** of the corresponding direction with respect to **classification performance**



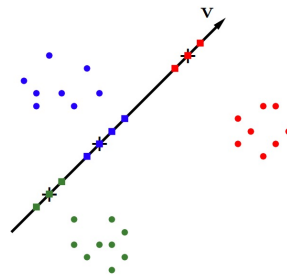
LDA

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Extension to Multiclass

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- Idea remains the same
 - **Tightness** of the projected classes is still described by the **total within-class scatter**



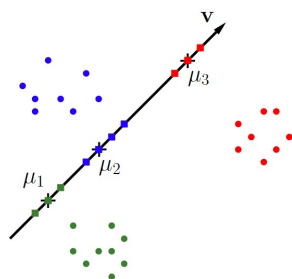
LDA

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Extension to Multiclass

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- Idea remains the same
- To maximize **between-class** separability - maximize the variance of the centroids



LDA

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Linear Discriminant Analysis

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- Linear Transformation (projection)

$$y = w^T x$$

- Fisher's criteria

$$J(w) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

Measure of between-class separability

Total within-class variance/scatter

$$|\tilde{m}_1 - \tilde{m}_2| = |w^T (m_1 - m_2)|$$

$$\tilde{s}_i^2 = \sum_{y \in y_i} (y - \tilde{m}_i)^2$$

LDA

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Linear Discriminant Analysis

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- Fisher's criteria as explicit function of w

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

The projection vector w is the eigenvector of $S_W^{-1} S_B$

LDA

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Linear Discriminant Analysis

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- The final step
 - Select the eigenvector that corresponds to the maximum eigenvalue to maximize class separability

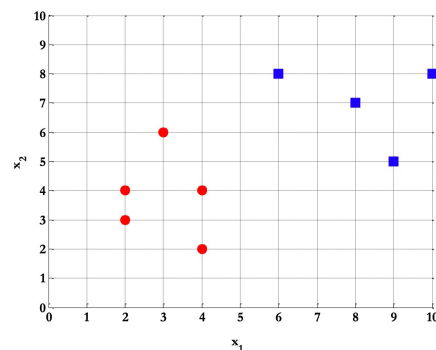
LDA

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LDA Example

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- Perform dimensionality reduction using LDA
- Dataset
 - Class 1: $\{(4,2), (2,4), (2,3), (3,6), (4,4)\}$
 - Class 2: $\{(9,10), (6,8), (9,5), (8,7), (10,8)\}$



LDA

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LDA Example

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- The class means are

$$\mu_1 = \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}$$

$$\mu_2 = \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix}$$

- Total within-class scatter matrix

$$\begin{aligned} S_w &= S_1 + S_2 \\ &= \begin{bmatrix} 0.8 & -0.2 \\ -0.2 & 1.76 \end{bmatrix} + \begin{bmatrix} 1.84 & -0.04 \\ -0.04 & 2.64 \end{bmatrix} = \begin{bmatrix} 2.64 & -0.24 \\ -0.24 & 4.4 \end{bmatrix} \end{aligned}$$

LDA

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LDA Example

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- Total within-class scatter matrix

$$S_w = \begin{bmatrix} 2.64 & -0.24 \\ -0.24 & 4.4 \end{bmatrix}$$

- Between-class scatter matrix

$$\begin{aligned} S_B &= (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \\ &= \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T \\ &= \begin{pmatrix} -5.4 \\ -3.8 \end{pmatrix} \begin{pmatrix} -5.4 & -3.8 \end{pmatrix} \\ S_B &= \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} \end{aligned}$$

LDA

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LDA Example

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- The LDA projection is then obtained as the solution of the following eigen value problem

$$S_W^{-1} S_B w = \lambda w$$

$$\begin{aligned} \square S_W^{-1} S_B &= \begin{bmatrix} 2.64 & -0.24 \\ -0.24 & 4.4 \end{bmatrix}^{-1} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} \\ &= \begin{bmatrix} 0.3807 & 0.0208 \\ 0.0208 & 0.2284 \end{bmatrix} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} \\ &= \begin{bmatrix} 11.5266 & 8.1113 \\ 5.2924 & 3.7243 \end{bmatrix} \end{aligned}$$

LDA

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LDA Example

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- The eigen values and eigen vectors of $S_W^{-1} S_B$

Eigenvalues

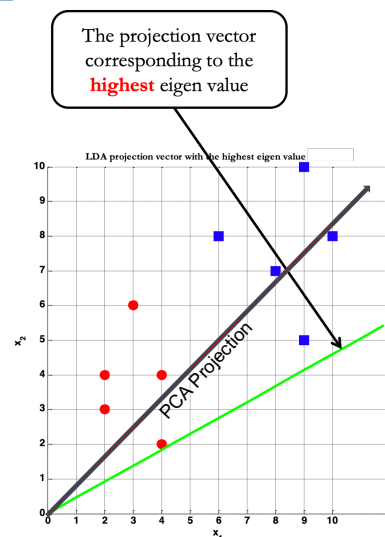
$$\lambda_1 \approx 15.2509$$

$$\lambda_2 \approx 0.0000178521$$

Eigenvectors

$$v_1 \approx (2.17795, 1)$$

$$v_2 \approx (-0.703704, 1)$$



LDA

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LDA Example

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- The eigen values and eigen vectors of $S_W^{-1} S_B$

Eigenvalues

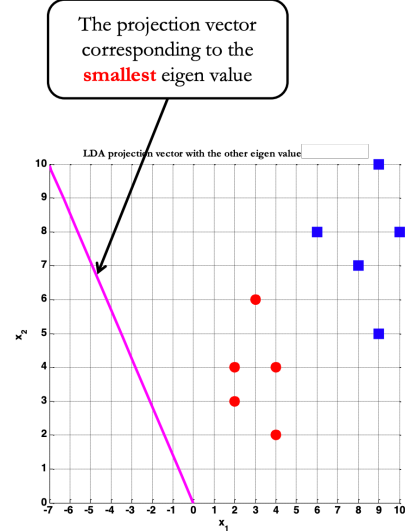
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Eigenvectors

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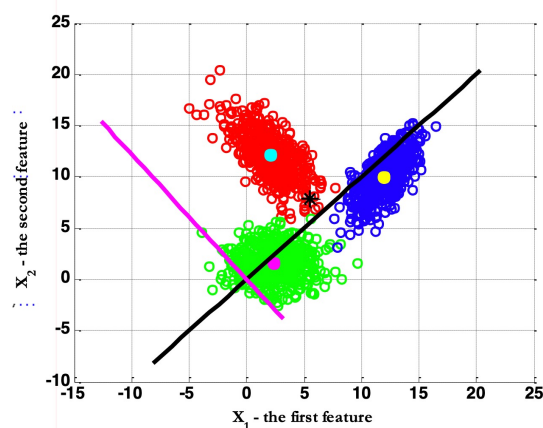


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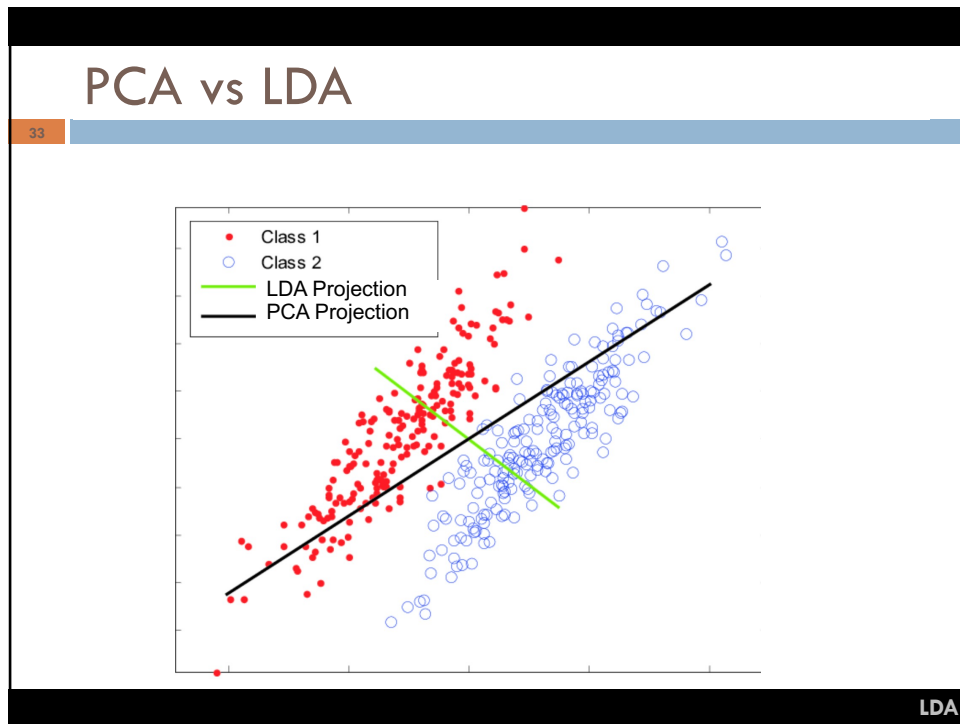
LDA Example

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LDA

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References

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- R. O. Duda, P. E. Hart and D. G. Stork, **Pattern Classification**. 2nd edition, Wiley-Interscience publication.
- A. Martinez, A. Kak, "**PCA versus LDA**", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 23, no. 2, pp. 228-233, 2001.

LDA

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Thank You!